



THE POWER OF WATER

Unchained, Unabated

NIDHI ARORA AND SANJAY BANERJEE

Koteshwar Power Project (Source: www.incoreinsightlytics.com)

'If the wars of the 20th Century were fought over oil, the wars of the 21st century will be fought over water—unless we change our approach to managing this precious and vital resource.'

—Ismail Serageldin

Hydroelectricity is one of the oldest and renewable forms of power generation. Of the total power produced in the world from renewable resources, about 70 per cent is generated through water. This makes up for about 16.6 per cent of the total electricity produced and is expected to increase at the rate of three per cent every year.

For hydroelectricity generation, the kinetic energy of flowing or falling water is used through gravitational force and converted into mechanical energy. This mechanical energy is used to turn turbines, which produces electricity.

As defined by an expert, “A hydraulic

turbine converts the energy of flowing water into mechanical energy. A hydroelectric generator converts this mechanical energy into electricity. The operation of a generator is based on the principles discovered by Faraday. He found that when a magnet is moved past a conductor, it causes electricity to flow. In a large generator, electromagnets are made by circulating direct current through loops of wire wound around stacks of magnetic steel laminations. These are called field poles and are mounted on the perimeter of the rotor. The rotor is attached to the turbine shaft and rotates at a fixed speed. When the rotor turns, it causes the field poles (electromagnets) to move past the conductors mounted in the stator. This, in turn, causes electricity to flow and a voltage to develop at the generator output terminals.”

Benefits of hydropower

1. Hydropower is produced using water—a clean fuel source. So unlike power plants that

